1.4: Fundamental Circuit Analysis

Kirchhoff's Current Law (KCL) and Voltage Law (KVL), Advanced Series/Parallel Analysis

1. Introduction to Circuit Analysis

Circuit analysis is the process of determining the voltages across and the currents through the components in an electrical circuit. Understanding fundamental laws like Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) is crucial for this analysis, especially when dealing with complex series and parallel configurations.

2. Kirchhoff's Current Law (KCL)

Kirchhoff's Current Law (KCL), also known as Kirchhoff's first law, states that the algebraic sum of currents entering a node (or a junction) in an electrical circuit is equal to the algebraic sum of currents leaving that node. In simpler terms, the total current entering a junction must be equal to the total current leaving it. This law is a direct consequence of the principle of conservation of charge.

Mathematically, KCL can be expressed as:

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

Or, equivalently:

$$\sum_{n=1}^{N} I_n = 0$$

Where I_n represents the n-th current entering or leaving the node. Currents entering the node are typically considered positive, and currents leaving are considered negative (or vice-versa, as long as consistency is maintained).

2.1. Example of KCL

Consider a node with three branches, where current I_1 enters, and currents I_2 and I_3 leave. If $I_1 = 5A$ and $I_2 = 2A$, we can find I_3 using KCL.

$$I_1 = I_2 + I_3$$

$$5A = 2A + I_3$$

$$I_3 = 5A - 2A = 3A$$

Therefore, I_3 is 3A leaving the node.

2.2. Practice Problem (KCL)

Problem: In a circuit, four currents meet at a node. Currents $I_A = 4A$ and $I_B = 6A$ enter the node. Current $I_C = 7A$ leaves the node. What is the magnitude and direction of the fourth current I_D?

Solution:

Using KCL:

$$I_A + I_B = I_C + I_D$$

$$4A + 6A = 7A + I_D$$

$$10A = 7A + I_D$$

$$I_D = 10A - 7A = 3A$$

Since I_D is positive, it means I_D is leaving the node.

3. Kirchhoff's Voltage Law (KVL)

Kirchhoff's Voltage Law (KVL), also known as Kirchhoff's second law, states that the algebraic sum of all voltages around any closed loop (or mesh) in an electrical circuit is equal to zero. This law is a direct consequence of the principle of conservation of energy.

Mathematically, KVL can be expressed as:

$$\sum_{n=1}^{N} V_n = 0$$

Where V_n represents the n-th voltage drop or rise in the loop. When traversing a loop, voltage drops (e.g., across a resistor in the direction of current) are typically considered negative, and voltage rises (e.g., across a voltage source from negative to positive terminal) are considered positive (or vice-versa, as long as consistency is maintained).

3.1. Example of KVL

Consider a simple series circuit with a voltage source $V_S = 12V$ and two resistors $R_1 = 2\Omega$ and $R_2 = 4\Omega$. We can find the voltage drops across each resistor, V_1 and V_2, using KVL.

Assuming a current I flows clockwise, the voltage drops are $V_1 = IR_1$ and $V_2 = IR_2$. By Ohm's Law, $I = V_S/(R_1 + R_2) = 12V/(2\Omega + 4\Omega) = 12V/6\Omega = 2A$.

Applying KVL around the loop:

$$V_S - V_1 - V_2 = 0$$

$$12V - (I \times R_1) - (I \times R_2) = 0$$

$$12V - (2A \times 2\Omega) - (2A \times 4\Omega) = 0$$

$$12V - 4V - 8V = 0$$

$$0 = 0$$

This confirms KVL.

3.2. Practice Problem (KVL)

Problem: A series circuit has a 20V voltage source, a 5Ω resistor, and an unknown resistor R_X. If the current flowing through the circuit is 2A, what is the value of R_X?

Solution:

Using Ohm's Law, the voltage drop across the 5Ω resistor is $V_1 = I \times R_1 = 2A \times 5\Omega = 10V$.

Applying KVL around the loop:

$$V_S - V_1 - V_X = 0$$

$$20V - 10V - (I \times R_X) = 0$$

$$10V - (2A \times R_X) = 0$$

$$2A \times R_X = 10V$$

$$R_X = 10V/2A = 5\Omega$$

Therefore, the value of the unknown resistor R_X is 5Ω.

4. Advanced Series/Parallel Analysis

Combining KCL and KVL with Ohm's Law allows for the analysis of more complex circuits involving both series and parallel combinations of resistors.

4.1. Series Circuits

In a **series circuit**, components are connected end-to-end, forming a single path for current flow. The key characteristics are:

- **Current:** The current is the same through all components.
- **Voltage:** The total voltage across the series combination is the sum of the individual voltage drops across each component (KVL).
- **Equivalent Resistance:** The total equivalent resistance (R_{eq}) is the sum of individual resistances.

$$R_{eq} = R_1 + R_2 + ... + R_n$$

4.2. Parallel Circuits

In a **parallel circuit**, components are connected across the same two points, providing multiple paths for current flow. The key characteristics are:

- **Voltage:** The voltage is the same across all components.
- **Current:** The total current entering the parallel combination is the sum of the individual currents through each branch (KCL).
- **Equivalent Resistance:** The reciprocal of the total equivalent resistance (R_{eq}) is the sum of the reciprocals of individual resistances.

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_n}$$

For two resistors in parallel, a common formula is:

$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$$

4.3. Comparison of Series and Parallel Circuits

Feature	Series Circuit	Parallel Circuit
Current	Same through all components	Divides among branches (sum of branch currents)
Voltage	Divides across components (sum of voltage drops)	Same across all components
Equivalent Resistance	$R_{eq} = R_1 + R_2 + ... + R_n$	$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + ... + \frac{1}{R_n}$
Adding more resistors	Increases total resistance	Decreases total resistance

4.4. Advanced Analysis Example (Series-Parallel Combination)

Consider a circuit with a 12V source, $R_1 = 3\Omega$ in series with a parallel combination of $R_2 = 6\Omega$ and $R_3 = 3\Omega$.

Step 1: Calculate the equivalent resistance of the parallel combination (R_p):

$$R_p = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{6\Omega \times 3\Omega}{6\Omega + 3\Omega} = \frac{18\Omega^2}{9\Omega} = 2\Omega$$

Step 2: Calculate the total equivalent resistance (R_{eq}) of the circuit:

Now, R_1 is in series with R_p.

$$R_{eq} = R_1 + R_p = 3\Omega + 2\Omega = 5\Omega$$

Step 3: Calculate the total current (I_{total}) from the source:

Using Ohm's Law:

$$I_{total} = \frac{V_{source}}{R_{eq}} = \frac{12V}{5\Omega} = 2.4A$$

Step 4: Calculate the voltage drop across R_1 (V_1):

$$V_1 = I_{total} \times R_1 = 2.4A \times 3\Omega = 7.2V$$

Step 5: Calculate the voltage across the parallel combination (V_p):

Using KVL for the main loop:

$$V_{source} - V_1 - V_p = 0$$

$$12V - 7.2V - V_p = 0$$

$$V_p = 12V - 7.2V = 4.8V$$

Alternatively, using Ohm's Law for the parallel equivalent:

$$V_p = I_{total} \times R_p = 2.4A \times 2\Omega = 4.8V$$

Step 6: Calculate the currents through R_2 (I_2) and R_3 (I_3):

Since R_2 and R_3 are in parallel, the voltage across them is V_p.

$$I_2 = \frac{V_p}{R_2} = \frac{4.8V}{6\Omega} = 0.8A$$

$$I_3 = \frac{V_p}{R_3} = \frac{4.8V}{3\Omega} = 1.6A$$

Step 7: Verify KCL at the node entering the parallel combination:

$$I_{total} = I_2 + I_3$$

$$2.4A = 0.8A + 1.6A$$

$$2.4A = 2.4A$$

This confirms KCL.

4.5. Practice Problem (Series-Parallel Combination)

Problem: A circuit has a 15V source. Resistor $R_1 = 4\Omega$ is in series with a parallel combination of $R_2 = 10\Omega$ and $R_3 = 15\Omega$. Find the total current from the source and the current through R_2.

Solution:

Step 1: Calculate the equivalent resistance of the parallel combination (R_p):

$$R_p = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{10\Omega \times 15\Omega}{10\Omega + 15\Omega} = \frac{150\Omega^2}{25\Omega} = 6\Omega$$

Step 2: Calculate the total equivalent resistance (R_{eq}) of the circuit:

$$R_{eq} = R_1 + R_p = 4\Omega + 6\Omega = 10\Omega$$

Step 3: Calculate the total current (I_{total}) from the source:

$$I_{total} = \frac{V_{source}}{R_{eq}} = \frac{15V}{10\Omega} = 1.5A$$

Step 4: Calculate the voltage across the parallel combination (V_p):

$$V_p = I_{total} \times R_p = 1.5A \times 6\Omega = 9V$$

Step 5: Calculate the current through R_2 (I_2):

$$I_2 = \frac{V_p}{R_2} = \frac{9V}{10\Omega} = 0.9A$$

Therefore, the total current from the source is 1.5A, and the current through R_2 is 0.9A.

5. Conclusion

Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) are fundamental principles in electrical circuit analysis. KCL, based on charge conservation, dictates that the sum of currents entering a node equals the sum of currents leaving it. KVL, based on energy conservation, states that the sum of voltages around any closed loop is zero. When combined with Ohm's Law, these laws provide a powerful framework for analyzing both simple and complex series and parallel circuits, enabling the determination of unknown currents and voltages throughout a circuit.

6. References

[1] https://www.allaboutcircuits.com/textbook/direct-current/chpt-6/kirchhoffs-voltage-law/ - Kirchhoff's Voltage Law [2] https://www.allaboutcircuits.com/textbook/direct-current/chpt-6/kirchhoffs-current-law/ - Kirchhoff's Current Law [3] https://www.electronics-tutorials.ws/dccircuits/dcp_4.html - Series and Parallel Resistors